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A procedure to form powder activated carbon into activated carbon monolith

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Abstract Activated carbon monolith was prepared using powder-activated carbon, methyl cellulose as organic binder, and bentonite as clay binder. The used organic binder contributed to the plasticity of powder-activated carbon for well extruding into well-shaped bodies, and the used clay binder contributed to the mechanical strength of the bodies. The bodies were characterized by determination of adsorption capacity, surface area, and mechanical strength (both compression and impact strengths). After that, it was endeavored to improve the mentioned properties by sieving powder-activated carbon into a uniform distribution of particle size. The obtained results illustrated that making the particle size uniform led to improving the bodies' properties significantly.

Keywords Activated carbon monolith · Organic binder · Clay binder · Surface area · Adsorption capacity · Mechanical strength

1 Introduction

Porous carbon materials are characterized by their highly developed internal surface area and their porosity. High surface area of activated carbon allows the carbons to adsorb large amount and various types of chemicals from gasses or liquids [1–3]. Activated carbons are of interest as adsorbent in many industries such as food processing, pharmaceuticals,

chemical, petroleum, mining and nuclear industries, and are used in the automobile, air conditioner or air purifier, water treatment, and vacuum manufacturing [4–7].

This diversity of applications depends on the surface properties and the shape of activated carbon. Activated carbons are classified into different types according to their sizes and shapes: powder, granular, activated carbon fiber, and shaped form (such as honeycomb structure). Each mentioned type of activated carbon has its specific applications in various industries [8–10]. One of the most important applications of the activated carbon monolith is hydrogen storage application. In addition, activated carbon monolith is typically used in columns or beds for gas and vapor systems as well as for processing a number of liquids. Such carbons have been used in canisters in automobiles through which gasoline tank and carburetor in order to adsorb the gasoline vapor prior to release to the environment [11]. Selection of a proper shape of activated carbon for a particular application is very important. The importance of using a specific granular form of activated carbon for improving ambient air was recently discussed in the work of Pinto et al. [12]. The shaped activated carbon has to possess both sufficient mechanical strength and proper adsorption capacity in order to be suitable for the applications [11].

Different approaches were used to obtain shaped activated carbon, such as, transform into activated carbon a carbonaceous material previously deposited on the surface of silica monoliths [13] or modify the normal activation procedure introducing a step where, by compression, pellets are obtained [14]. If after the activation step no washing is needed, i.e., when physical activation is used, the last methodology can be an option to prepare granular carbons. Nevertheless, in the case of chemical activation the washing step is indispensable to leach out the activating agent to obtain an unobstructed porous system. This washing step usually leads to the

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destruction of the granular forms and so another strategy has to be followed, namely, compaction of activated carbons with binders [9].

In the research works, various binders were used, such as, humic acid and its sodium salt [14, 15]; organic binder [10, 15]; tar [15–17]; and clays [9, 18–21]. A plasticizing organic binder contributes to the plasticity of activated carbon for well shaping into a body [22]. Also high levels of organic binders (up to 50 wt%) are needed to achieve reasonable mechanical properties of the body. On the other hand, it lead considerably to decreasing of adsorption capacity of the shaped bodies as well as increasing of plasticized mixture viscosity and subsequently the impossibility of extrusion [10, 21]. The clay binder contributes mainly mechanical strength [23]. In a number of applications, the use of inorganic binders, such as clays, has advantageous due to their inertness and thermal resistance [9]. However, in such works, mixture's plasticity depends on the proportion of carbon/clay [9]. Therefore, preparing a mixture with proper plasticity (suitable proportion of carbon/clay) for well shaping is kind of challenging. To our knowledge, preparation of activated carbon monolith using a composition of a clay binder and an organic binder have not been studied yet.

In this work, a procedure was proposed to prepare activated carbon monolith by extruding a mixture of powder activated carbon, organic binder and clay binder into cylindrical bodies. The organic binder contributed to the plasticity of powder activated carbon for well extruding into a body and the clay binder contributed the mechanical strength of the bodies. Influence of the binders on the shaped bodies' textural, adsorption capacity, and mechanical properties (both compression and impact strength) were studied and compared. In addition, it was tried to improve the properties by screening the powder-activated carbon into uniform particle size.

2 Experimental

2.1 Materials and reagents

Methyl cellulose (BDH 400cps) and bentonite (shah industrial mining company) were used as organic and clay binders,

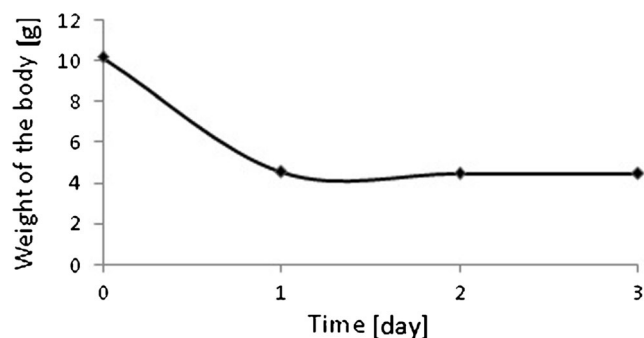


Fig. 1 Weight of the body against time

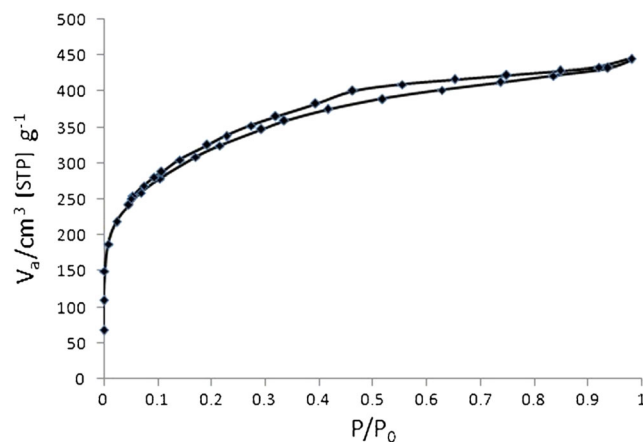


Fig. 2 Nitrogen adsorption-desorption curve related to the prepared powder activated carbon

respectively. The methyl cellulose has a gelation temperature (around 45 °C), and below the gelation temperature, the methyl cellulose is completely soluble in water. Bentonite chemically is alumina rich and has an alumina or magnesium sheet surrounded on either side by a silica sheet. Bentonite clay is a high-surface area material which is also porous [24]. Specific surface area of the bentonite was determined up to 16 m²/g. Chloridric acid (HCl), iodine (I), potassium iodide (KI), zinc chloride (ZnCl₂), starch, and sodium thiosulfate (Na₂S₂O₃) were obtained from Merck Company (Germany), all with mass fraction purity greater than 0.99. All chemicals were used without any further purification.

2.2 Preparation of powder-activated carbon and screening

Hazelnut shell (HS) was used as the starting material in preparation of powder-activated carbon. HS was first air-dried, crushed, and sized, the fraction of particle size

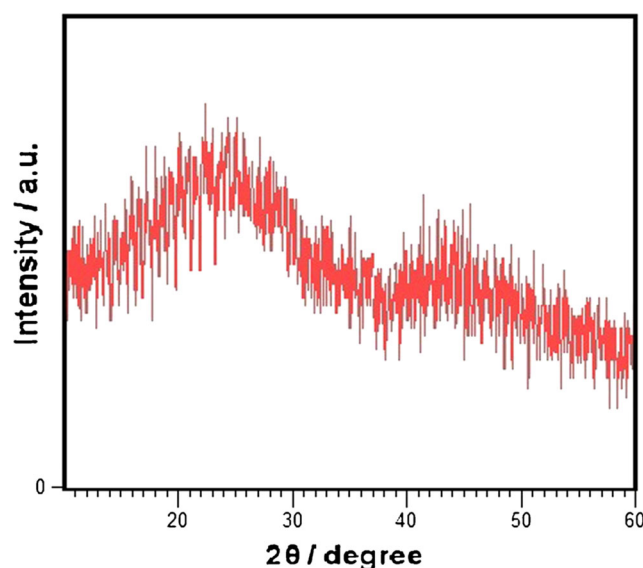


Fig. 3 X-Ray diffraction pattern of the prepared activated carbon

Table 1 Minimum content of methyl cellulose for producing a body having good appearance shape and strength

Number of test	Bentonite wt%	Methylcellulose wt%	Quality of appearance of the bodies	Strength (MPa)
1	10	1	Poor	—
2	10	2	Poor	—
3	10	3	Poor	—
4	10	4	Poor	—
5	10	5	Medium	1.7
6	10	6	Good	3.5
7	10	8	Good	3.6

between 0.1 to 1 mm being chosen. In the impregnation of HS, 150 ml of ZnCl_2 solution were brought into contact with a certain amount of HS (ZnCl_2 to HS weight ratio was 4) at 80 °C for 8 h under mechanical agitation. Evaporation during this time was avoided using a close-up glass container. Once such a time elapsed, the supernatant liquid was separated by vacuum filtration and the remaining solid was oven-dried at 100 °C for 24 h.

The impregnated sample was placed on a quartz dish, which was then inserted in a quartz tube. The impregnated sample was heated up to activation temperature (400 °C) under N_2 flow at the rate of 10 °C min^{-1} and hold at the activation temperature for 1 h. After activation, the sample was cooled down under N_2 flow. The resulting solids after carbonization were boiled at about 90 °C with 100 ml of 1 M HCl solution for 30 min to leach out the activating agent, filtered and rinsed by warm distilled water several times to remove excess acid. They were then dried at 110 °C for 12 h.

2.3 Manufacture of the activated carbon monolith

A mixture of powder-activated carbon, bentonite, methylcellulose, and water, which have certain weight percent (wt%), was prepared. The mixture was agglomerated in a laboratory mixer with 300 rpm at 60 °C for 40 min. The produced semi-plastic mixture was passed through a laboratory extruder with a 1-cm-diameter hole. The extruded product was then reduced to cylinders having 1 cm long. The required time to

completely remove water from the shaped bodies was determined. For this purpose, a body having certain weight (10.77 g) was prepared. Then, the body was placed in an oven at 100 °C. Weight loss of the body was measured by day to day. Figure 1 shows weight of the body against day. As can be seen in Fig. 1, weight of the body after third day did not change. Therefore, in this work, the bodies were dried in three days at temperature 100 °C.

In the above mentioned step for preparation of the semi-plastic mixture, the temperature was regulated by a thermostatic bath (Grant instruments (Cambridge) Ltd. (type SE 15, England)), accurate to ± 0.2 °C. Also, the components were weighed on an analytical balance (A&D Company, GF-600, Japan) accurate to 0.001 g. It is important to note that the mixing of a binder with powder-activated carbon using high speed mixer leads to prevent the blocking of the activated carbon pores and improves adsorption capacity of the prepared mixture [25].

2.4 Measurement of the properties

Appearance of the bodies was examined by taking photos using Canon (EOS 60D) camera. The iodine numbers of the samples were measured based on the ASTM D4607–94 [26]. In this work, all the iodine number measurements were performed three times and the average value was reported. The BET surface area was measured using Belsorp mini II apparatus (BelJapan). The compression strength was measured as a body was destroyed for the first time by supplying a force onto the body using an U.T.M. device (Dartec 9500) at a velocity up to 10 mm/min. The impact strength of the bodies was tested quantifying the weight loss after 30 drops from a height of 55 cm on a steel plate. The results correspond to the mean of ten essays. The procedure was reported in reference [9]. The crystalline structures of the product were examined by an X-ray diffractometer (Siemens, Model D-500) using $\text{Cu-K}\alpha$ radiation with a scan speed of 2.5°min^{-1} . The powder-activated carbon was sieved by standard sieves equipped with a shaker (E.L.M., SIE. 35). Also, the SEM images were recorded using XL30 Philips.

Fig. 4 Images of the produced bodies containing various contents of methyl cellulose. **a** 4 wt% methylcellulose (poor appearance), **b** 5 wt% methylcellulose (medium appearance), **c** 6 wt% methylcellulose (good appearance)

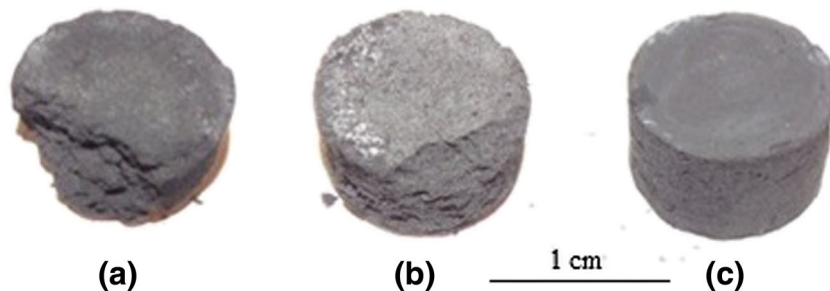


Table 2 Iodine number, surface area, and mechanical strength of the produced bodies

Number of test	Bentonite wt%	Methylcellulose wt%	Specific surface area (m ² /g)	Iodine number (mg/g)	Compression strength (MPa)	Weight loss after the impact strength (%)
1	10	6	1095	1290.99	3.5	5.96
2	20	6	1005	1238.14	3.7	5.87
3	25	6	940	1222.41	4.2	4.36
4	30	6	902	1194.12	4.5	3.28
5	35	6	825	1185.81	4.8	3.01
6	40	6	775	1169.43	4	5.26

3 Result and discussion

3.1 Characteristics of the prepared activated carbon

The measured iodine number of the prepared activated carbon based on the ASTM D4607–94 is about 1380 mg/g. Specific surface area of the activated carbon was measured up to 1163 m²/g. Figure 2 shows N₂ adsorption-desorption isotherms curve related to the prepared powder activated carbon. The prepared activated carbon presents a modified type I isotherm, corresponding to a well-developed microporous structure with a significant contribution of mesoporosity.

Figure 3 shows the XRD pattern of the activated carbon obtained from HS and ZnCl₂. As can be seen in Fig. 3, two broad peaks appear at approximately 2 θ =25 and 45° after activation with ZnCl₂, indicating the formation of turbostratic structure of disordered carbon. Kennedy et al. [27] reported that peaks at 2 θ =45° are due to the creation of pores by the decomposition of carbon along the direction of the graphic structures. This produces relatively well-organized aromatic carbon that is more stable than amorphous-like carbon.

3.2 Minimizing methyl cellulose content

It is clear that the surface area depends on amount of the organic binder contained in the initial mixture. Increasing

organic binder content leads to decreasing surface area [18]. Therefore, it is crucial to minimize content of methylcellulose in the initial mixture. For this purpose, the minimum content of methyl cellulose was determined to reach appropriate shape and strength. After that, mechanical strength was improved by increasing bentonite content. As can be seen in Table 1, bentonite quantity in the mixture remained constant up to 10 wt%. Also, content of methyl cellulose was increased from 1 wt% onward, attaining an appropriate amount. The produced body contained 1 to 5 wt% methyl cellulose did not come up with appropriate quality as appearance shape and strength. A rise in methyl cellulose content to 6 wt% led to producing a body having a good quality as both appearance shape and strength. Figure 4 shows images of the produced bodies contained various content of methyl cellulose. As can be seen in Fig. 4, the produced bodies having 4 and 5 wt% methyl cellulose did not come up with proper appearance shape. However, the produced body containing 6 wt% methyl cellulose had suitable shape without any cracks on the surface.

3.3 Effect of bentonite and methyl cellulose contents on mechanical strength, adsorption capacity, and surface area of the bodies

As mentioned earlier, after minimizing methyl cellulose content, it was tried to boost the mechanical strength of

Table 3 Iodine number, surface area, and mechanical strength of the produced bodies with increased amount of methyl cellulose

Number of test	Bentonite wt%	Methylcellulose wt%	Specific surface area (m ² /g)	Iodine number (mg/g)	Compression strength (MPa)	Weight loss after the impact strength (%)
1	10	8	1025	1285.93	3.6	5.57
2	20	8	980	1235.84	3.73	3.88
3	25	8	915	1209.92	4.3	3.17
4	30	8	970	1191.59	4.54	2.13
5	35	8	790	1162.86	5.1	1.22
6	40	8	735	1152.03	4.5	2.25

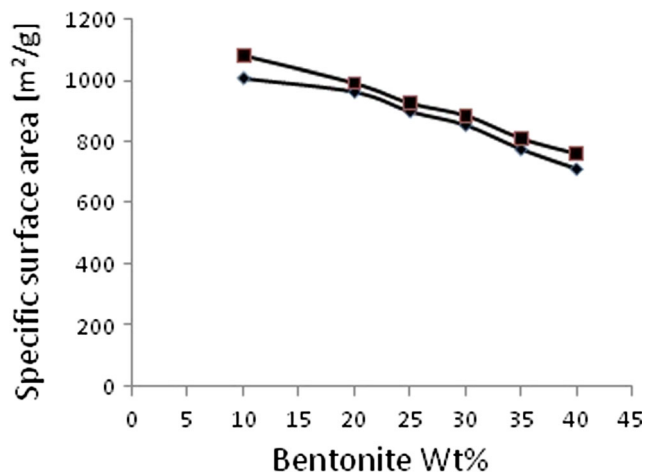


Fig. 5 A comparison between surface area of the bodies containing 6 wt% (black square) and 8 wt% (black diamond) methyl cellulose (MC) and various contents of bentonite

the bodies by increasing the bentonite content. As can be seen in Table 2, the methyl cellulose content stayed constant up to the minimum proper content (6 wt%), and then bentonite content was increased from 10 to 40 wt%, resulting in improvement of the compression and impact strength. It is worth noting that increasing bentonite content did not cause a significant drop in the adsorption capacity and surface area.

Increasing methyl cellulose content also would result in improvement of mechanical strength. However, it would cause a significant dip in the adsorption capacity. In addition, the shaping process would hardly perform and need a careful drying process. Also, it was investigated whether adding more methyl cellulose content would impact on mechanical strength and adsorption capacity. As can be seen in Table 3, the methyl cellulose content was remained

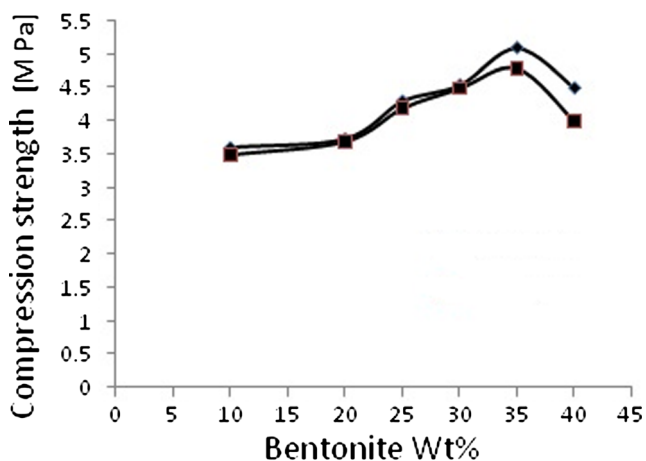


Fig. 6 A comparison between compression strength of the bodies containing 6 wt% (black square) and 8 wt% (black diamond) methyl cellulose (MC) and various contents of bentonite

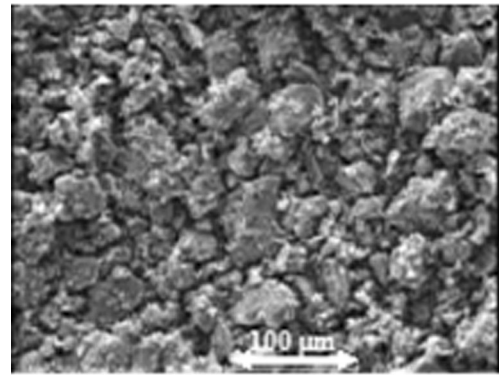


Fig. 7 SEM image of powder-activated carbon with particle size less than 90 μm (250 $\times$)

constant up to 8 wt% and the bentonite content was increased from 10 to 40 wt%. In Fig. 5, the specific surface areas regarding the bodies with 6 and 8 wt% methyl cellulose and various contents of bentonite (10–40 wt%) were compared. As can be seen in Fig. 5, increasing methyl cellulose content from 6 to 8 wt% led to decreasing the surface areas. Therefore, a slight increase in methyl cellulose content resulted in a significant negative influence on the surface area. Besides, the obtained iodine number of the bodies confirmed that increasing the methylcellulose content effected on adsorption capacity negatively (Tables 2 and 3).

Figure 6 shows a comparison between mechanical strength of the bodies containing 6 and 8 wt% methyl cellulose and various contents of bentonite. It is clear that rising methyl cellulose content resulted in increasing the mechanical strength.

As can be seen in Fig. 6 and Tables 2 and 3, increasing bentonite content through 35 wt% led to rising the compression and impact strength. As the bentonite content was increased for more 5 wt%, the compression and impact strength were suddenly decreased. Bentonite has a unique characteristic, and when it mixes with water or vapor, it swells [24]. The mentioned decrease can be contributed to this unique characteristic of the bentonite clay. Therefore, it can be mentioned that increasing bentonite content till 40 wt% led to further swelling, resulting in drop in the bodies' properties.

3.4 Improvement of the produced bodies' properties by screening the powder activated carbon into uniform particle size

The powder activated carbon was sieved to obtain powder activated carbon with particles less than 90 μm . Figure 7 shows SEM image of the sieved powder activated carbon. Then, the sieved powder activated carbon was

Table 4 Properties of the bodies produced by powder activated carbon with particle size 150 μm

Sample	Bentonite wt%	Methylcellulose wt%	Powder activated carbon particle size (μm)	Specific surface area (m^2/g)	Iodine number (mg/g)	Compression strength (MPa)	Weight loss after the impact strength (%)
1	20	6	90	1085	1290.65	4	4.56
2	20	6	Various ^a	1005	1238.14	3.7	5.87
3	20	8	90	994	1247.31	4.1	3.36
4	20	8	Various ^a	980	1235.84	3.73	3.88

^a Without screening

used to produce new bodies. Table 4 shows iodine number, surface area, and mechanical strength of the bodies manufactured by powder-activated carbon with uniform particle size. As can be seen in Table 4, this sieving resulted in a rise about 52 mg/g in iodine number. Such improvement was experienced in specific surface area and mechanical strength (both compression and impact strength) by the bodies.

3.5 Comparison of the produced bodies in this work with other works

Table 5 shows a comparison between the produced bodies in this work and other works. Samples 1 and 2, which were chosen for this comparison, constituted the highest mechanical strength and the highest adsorption capacity between all the bodies produced here, respectively. As can be

Table 5 Comparison between the produced bodies in this work and other works

Sample	BET (m^2/g)	Compression strength (Mpa)	Weight loss after impact strength test (%)	Iodine number (mg/g)	Reference
1	790	5.1	1.22	1162	Present work
2	1095	3.5	5.96	1290	Present work
3	799	—	—	495	[6]
4	802	—	5.4	—	[9]
5	774	—	4.6	—	[9]
6	564	0.1	—	—	[10]
7	347	—	—	—	[13]
22	521	—	—	—	[17]
8	710	—	—	—	[18]
9	622	—	—	—	[18]
10	633	—	—	—	[19]
11	733	—	—	—	[19]
12	853	—	—	—	[28]
13	750	—	—	—	[28]
14	820	—	—	—	[29]
15	804	—	—	—	[29]
16	546	—	—	—	[30]
17	710	—	—	—	[31]
18	45	—	—	—	[32]
19	473	—	—	—	[33]
20	469	—	—	—	[34]
21	487	—	—	—	[35]
23	412	2.8	—	—	[36]
24	425	—	—	—	[37]
25	805	—	—	—	[38]
26	715	4.8	—	—	[39]
27	710	4.2	—	—	[39]
28	624	—	—	—	[40]

seen in Table 5, compared to other reported works, sample 2 accounted for the highest iodine number and surface area. In addition, the highest mechanical strength belonged to the sample 1.

4 Conclusion

In this work, well-shaped activated carbon monolith with high surface area, good adsorption capacity, and proper mechanical strength was prepared. The way, in which clay and organic binders were simultaneously used, led to preparing a suitable mixture having proper ability for well-extruding process, resulting in well-shaped bodies without any cracks.

The obtained results related to surface area, adsorption capacity, and mechanical strength (both compression and impact strength) were compared with the results reported in the literature. This comparison proved that the prepared bodies have better qualities than that of other works. It is also worth pointing out that screening powder-activated carbon into uniform particle size led to further improving adsorption capacity, surface area, and mechanical strength of the produced activated carbon monolith.

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